

# AI Smart Stethoscope -- Diagnosis Report

Kuwait College of Science & Technology -- Group 07

AI-generated report. Does not replace professional medical diagnosis.

## PATIENT & SESSION INFORMATION

|                   |                                 |
|-------------------|---------------------------------|
| Patient ID:       | DEMO-PATIENT-001                |
| Date & Time:      | 2026-07-28 14:10:11             |
| Recording Mode:   | DEMO                            |
| Source File:      | normal__128_1306344005749_D.wav |
| Diagnosis:        | Extra Heart Sound (S3/S4)       |
| Murmur Prob:      | 1.1%                            |
| Murmur Threshold: | 0.50 (conservative for safety)  |
| Est. Heart Rate:  | 100 BPM                         |

## CLASS PROBABILITIES

|          |             |       |
|----------|-------------|-------|
| ARTIFACT | <div></div> | 0.4%  |
| EXTRAHLS | <div></div> | 55.5% |
| MURMUR   | <div></div> | 1.1%  |
| NORMAL   | <div></div> | 43.0% |

## CLINICAL FINDINGS

- Extra low-frequency component detected after S2 (possible S3)
- Triple rhythm (gallop) pattern identified in waveform
- Energy burst outside normal S1-S2 cardiac cycle
- Pattern consistent with ventricular filling abnormality
- Mel spectrogram shows additional energy cluster post-S2

## DETAILED CLINICAL REASONING

The AI model identified an additional acoustic event beyond the expected S1-S2 cardiac cycle, classified as an extra heart sound (S3 or S4 gallop). An S3 may indicate heart failure, dilated cardiomyopathy, or volume overload; an S4 suggests reduced ventricular compliance or hypertensive heart disease. Physiological S3 can be normal in athletes and patients under 40, so clinical context is essential. The extra-sound probability score (1.1%) indicates a clear departure from the normal two-sound pattern. Clinical correlation is mandatory.

## MEDICAL RECOMMENDATION

Cardiac evaluation advised. Consider BNP/NT-proBNP levels and

echocardiography for a suspected S3; evaluate for hypertension or hypertrophic cardiomyopathy for a suspected S4. Refer to a cardiologist for comprehensive hemodynamic assessment.

# Signal Analysis -- Waveforms & Spectrograms

